```python
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("/Users/moshika/Downloads/techmart_sales.csv")
df['Order_Date'] = pd.to_datetime(df['Order_Date'])
df['Year'] = df['Order_Date'].dt.year
df['Month'] = df['Order_Date'].dt.month
df['Quarter'] = df['Order_Date'].dt.quarter
df['Revenue_after_discount'] = df['Sales'] * (1 - df['Discount'])
df['Profit_Margin'] = df['Profit'] / df['Sales']
monthly_sales = df.groupby('Month')['Sales'].sum()
plt.figure()
monthly_sales.plot()
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.show()
category_sales = df.groupby('Category')['Sales'].sum()
plt.figure()
category_sales.plot(kind='bar')
plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.show()
regional_profit = df.groupby('Region')['Profit'].sum()
plt.figure()
regional_profit.plot(kind='bar')
plt.title("Total Profit by Region")
plt.xlabel("Region")
plt.ylabel("Total Profit")
plt.xticks(rotation=0)
plt.show()
```

```python
bins = [0, 0.1, 0.2, 0.3, 1]
labels = ['0-10%', '10-20%', '20-30%', '>30%']
df['Discount_Range'] = pd.cut(df['Discount'], bins=bins, labels=labels)
discount_profit = df.groupby('Discount_Range')['Profit'].sum()
plt.figure()
discount_profit.plot(kind='bar')
plt.title("Average Profit by Discount Range")
plt.xlabel("Discount Range")
plt.ylabel("Average Profit")
plt.xticks(rotation=0)
plt.show()
payment_sales = df.groupby('Payment_Mode')['Sales'].sum()
plt.figure()
payment_sales.plot(kind='bar')
plt.title("Sales by Payment Mode")
plt.xlabel("Payment Mode")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.show()
```

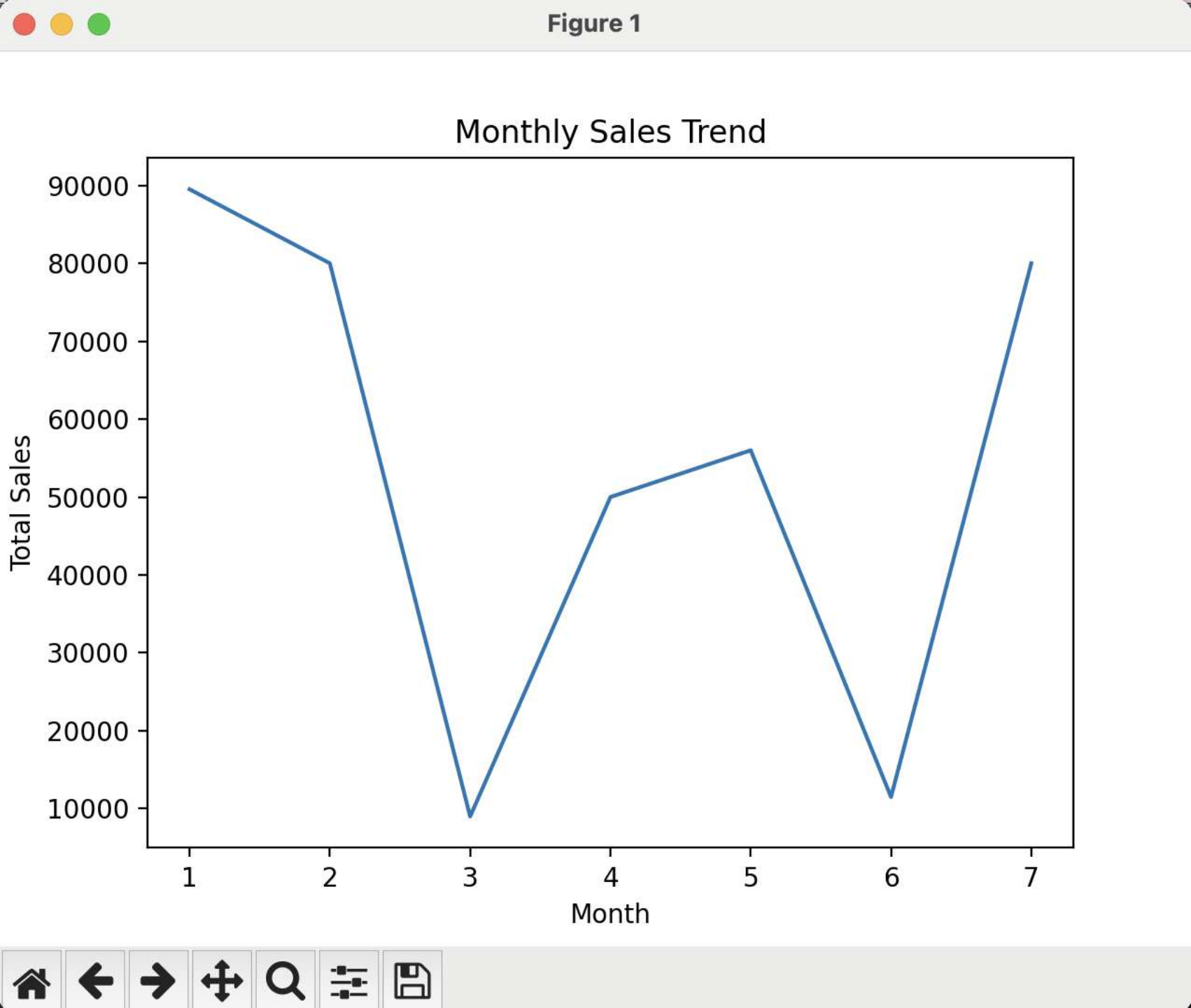
Figure 1
Monthly Sales Trend
Total Sales
90000
80000
70000
60000
50000
40000
30000
20000
10000
1
2
3
4
5
6
7
Month

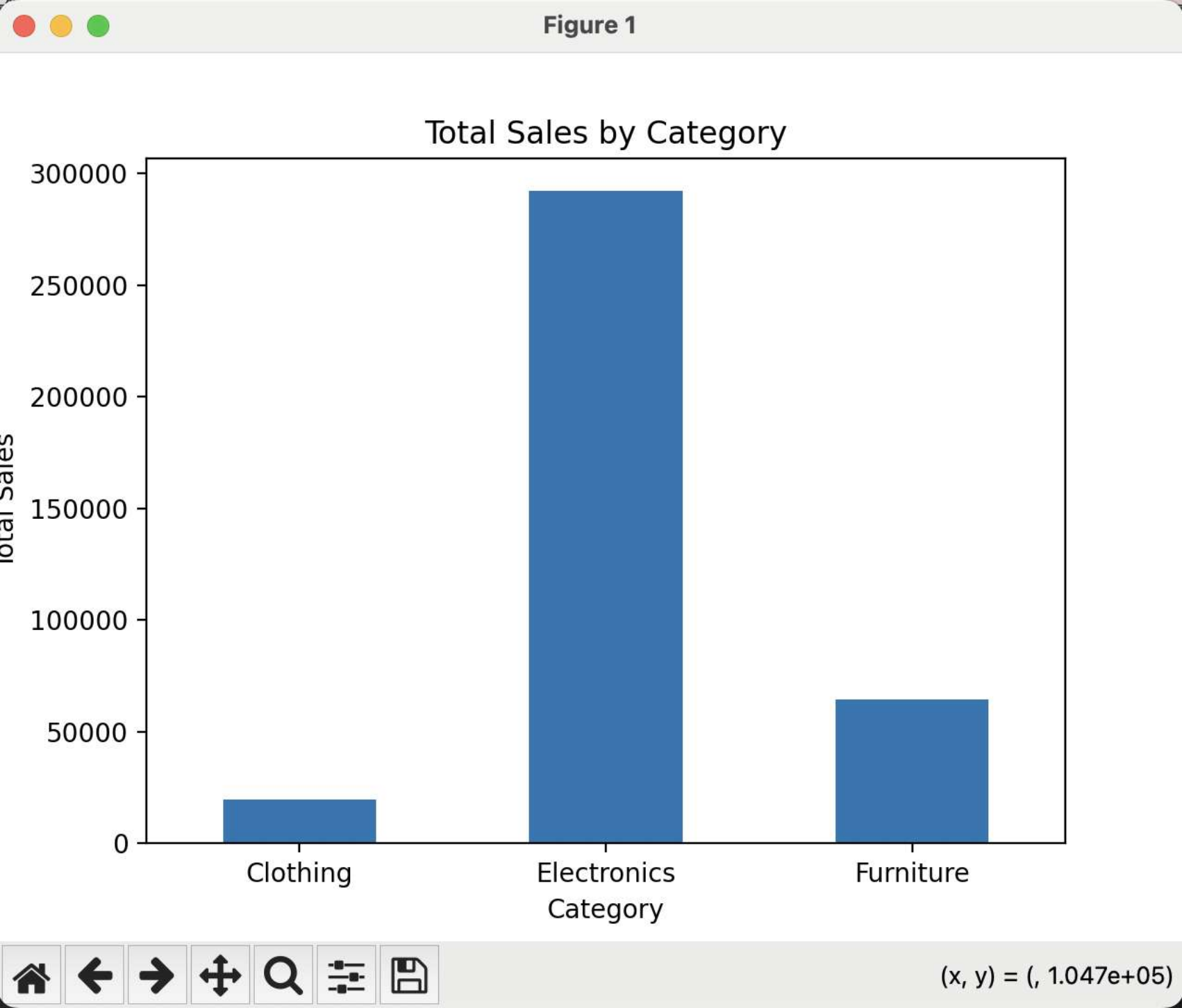
Figure 1
Total Sales by Category
300000
250000
200000
150000
100000
50000
0
Total Sales
Clothing
Electronics
Furniture
Category
(x, y) = (, 1.047e+05)

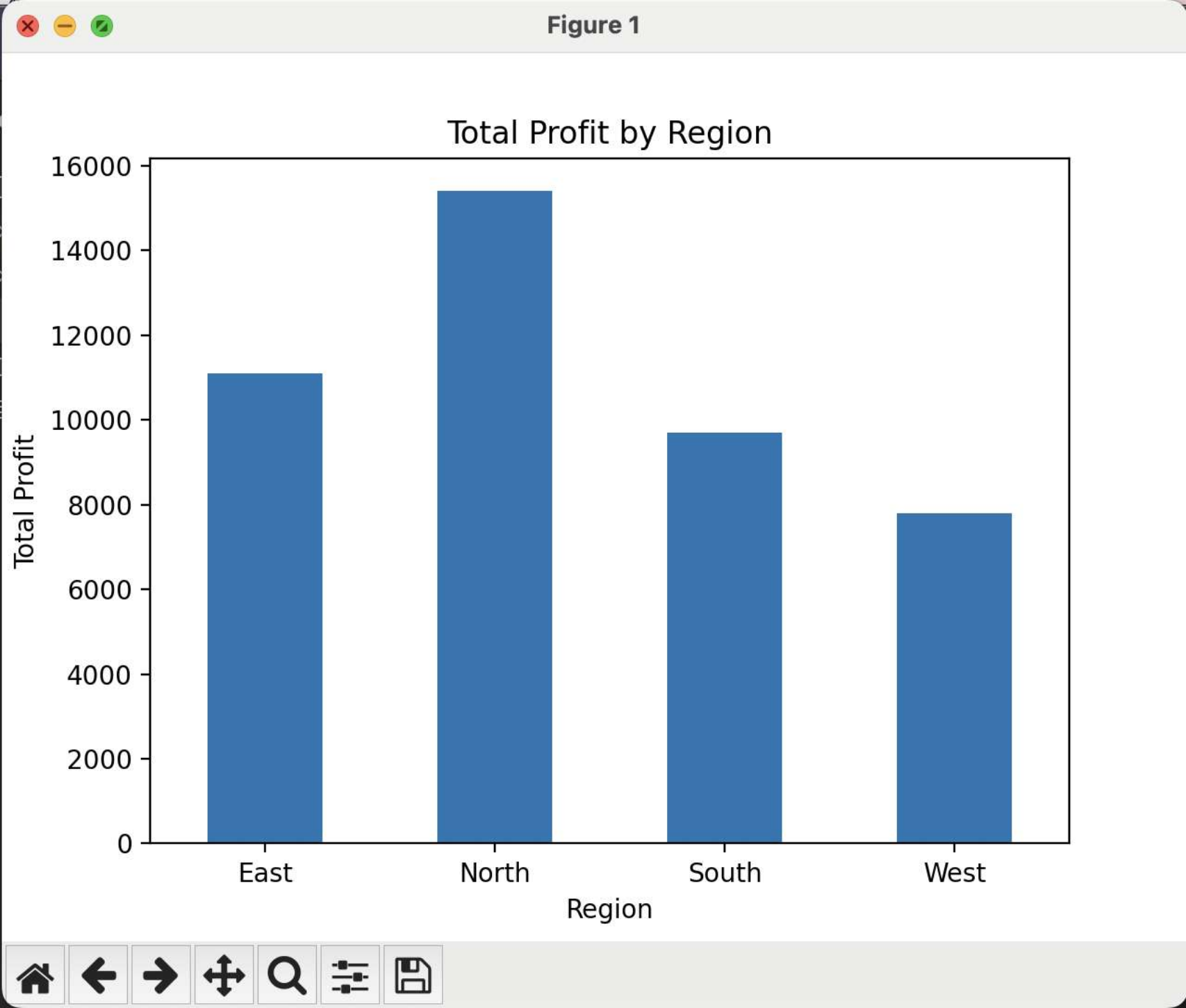
Figure 1
Total Profit by Region
16000
14000
12000
10000
8000
6000
4000
2000
0
Total Profit
East
North
South
West
Region

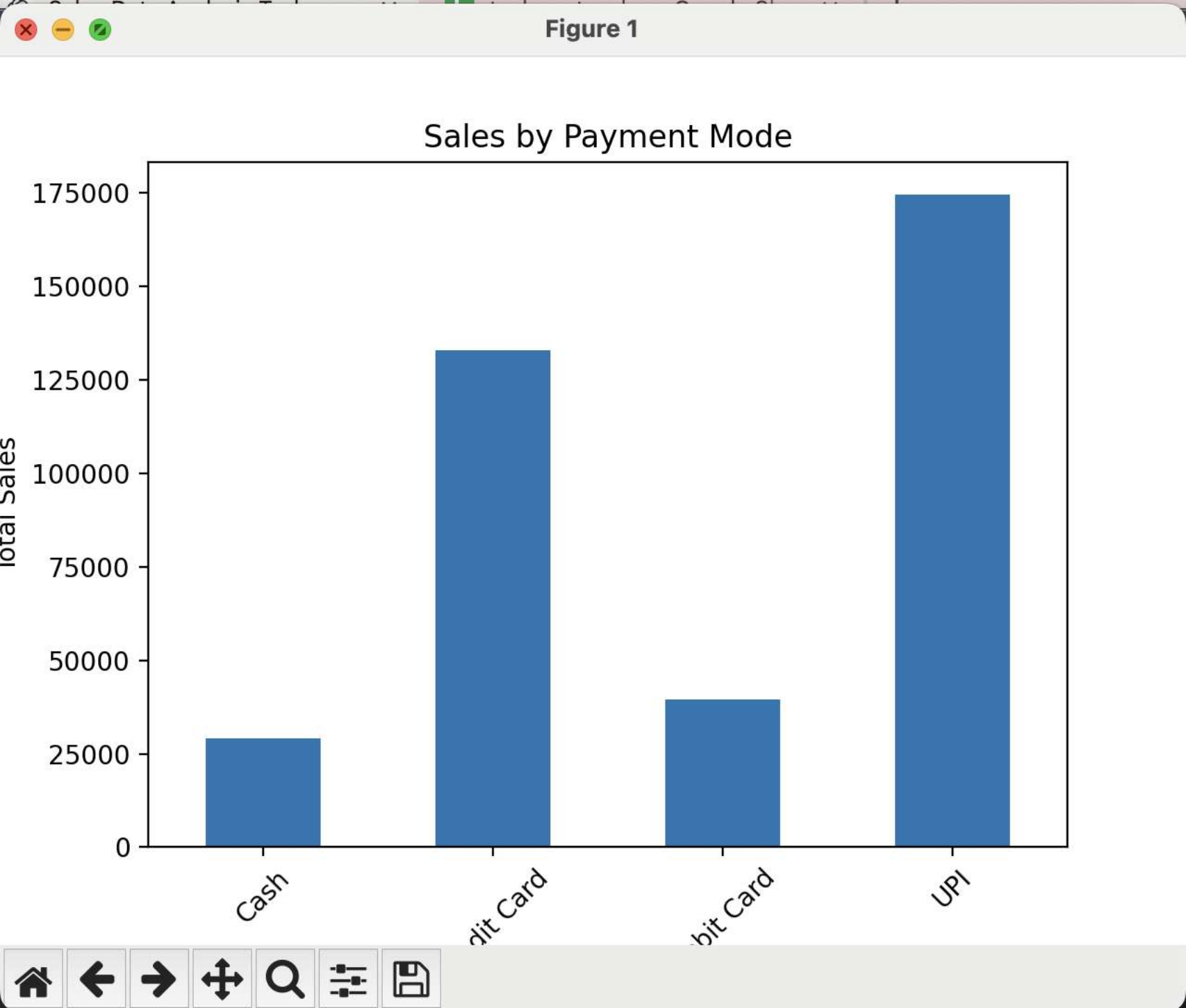
Figure 1
Sales by Payment Mode
175000
150000
125000
100000
75000
50000
25000
0
Cash
UPI

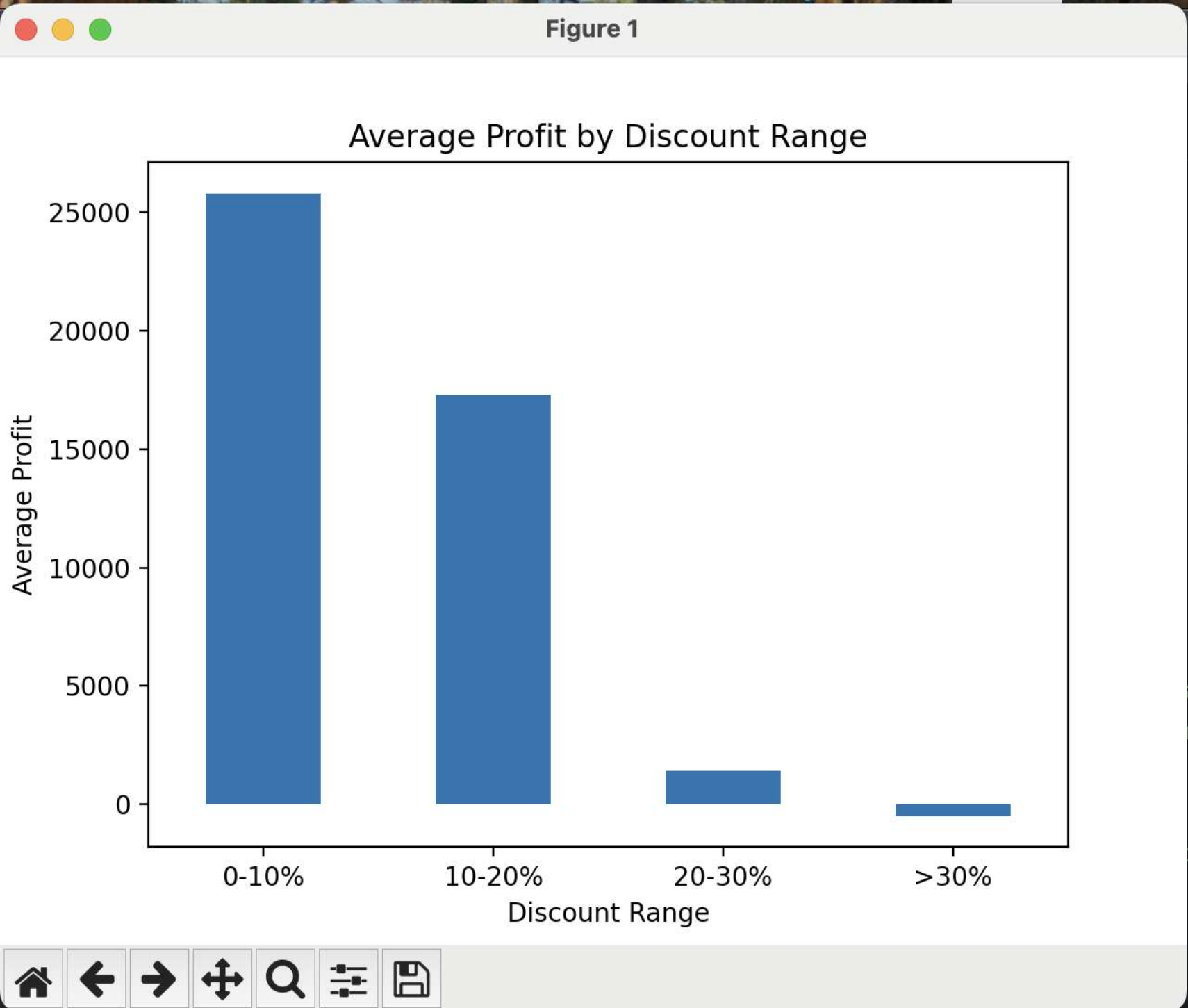
Figure 1
Average Profit by Discount Range
Average Profit
25000
20000
15000
10000
5000
0
0-10%
10-20%
20-30%
>30%
Discount Range